Amrutvahini College of Engineering, Sangamner

Department of Electronics and Telecommunication Engineering

Experiment No.: 07

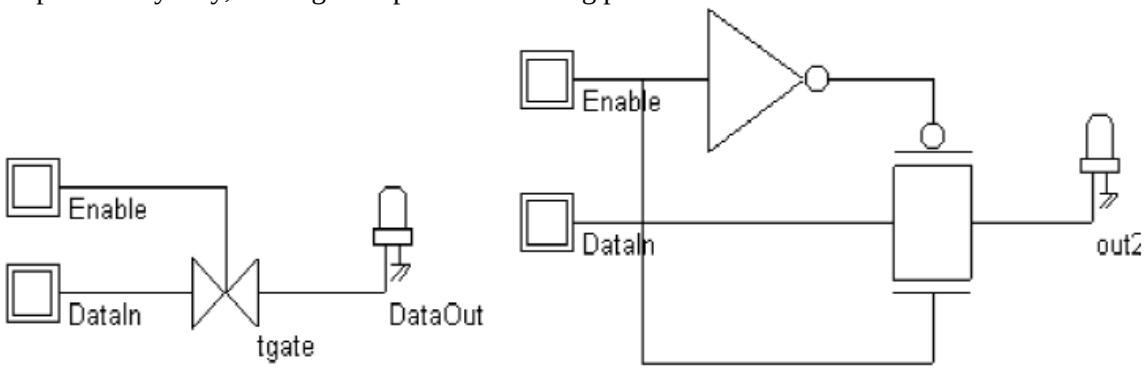
Date of Performance:

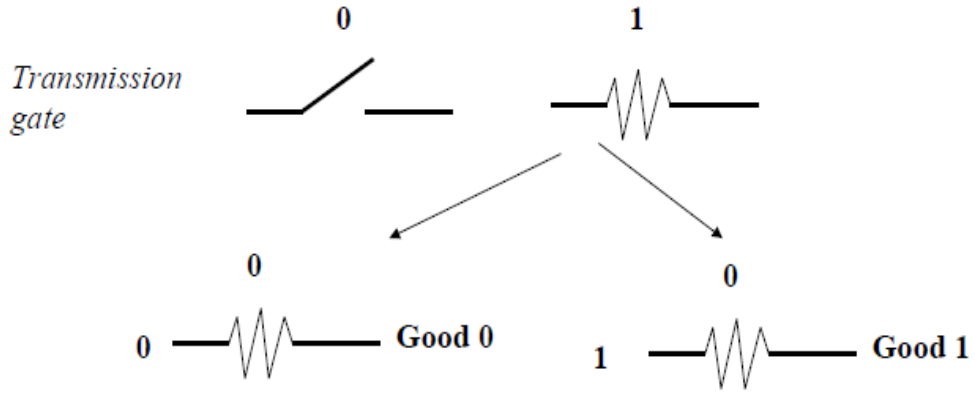
AIM: - Design of 2:1 Mux by conventional method and by using Transmission gates and Comparison of them

OBJECTIVE: - To study importance of transmission gate.

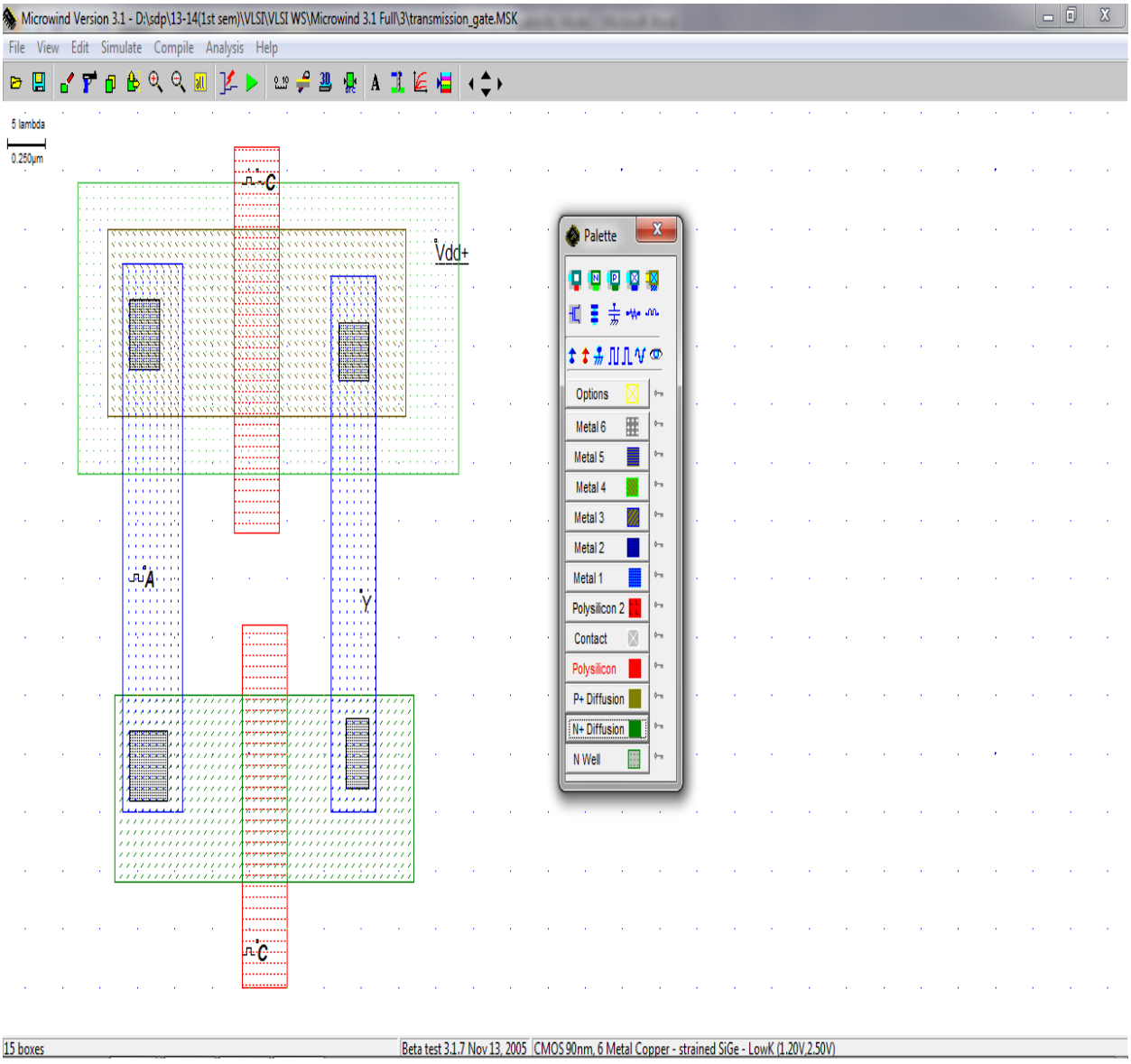
THEORY: -

Both NMOS devices and PMOS devices exhibit poor performances when transmitting one particular logic information. The nMOS degrades the logic level 1, the pMOS degrades the logic level 0. Thus, a perfect pass gate can be constructed from the combination of nMOS and pMOS devices working in a complementary way, leading to improved switching performances.

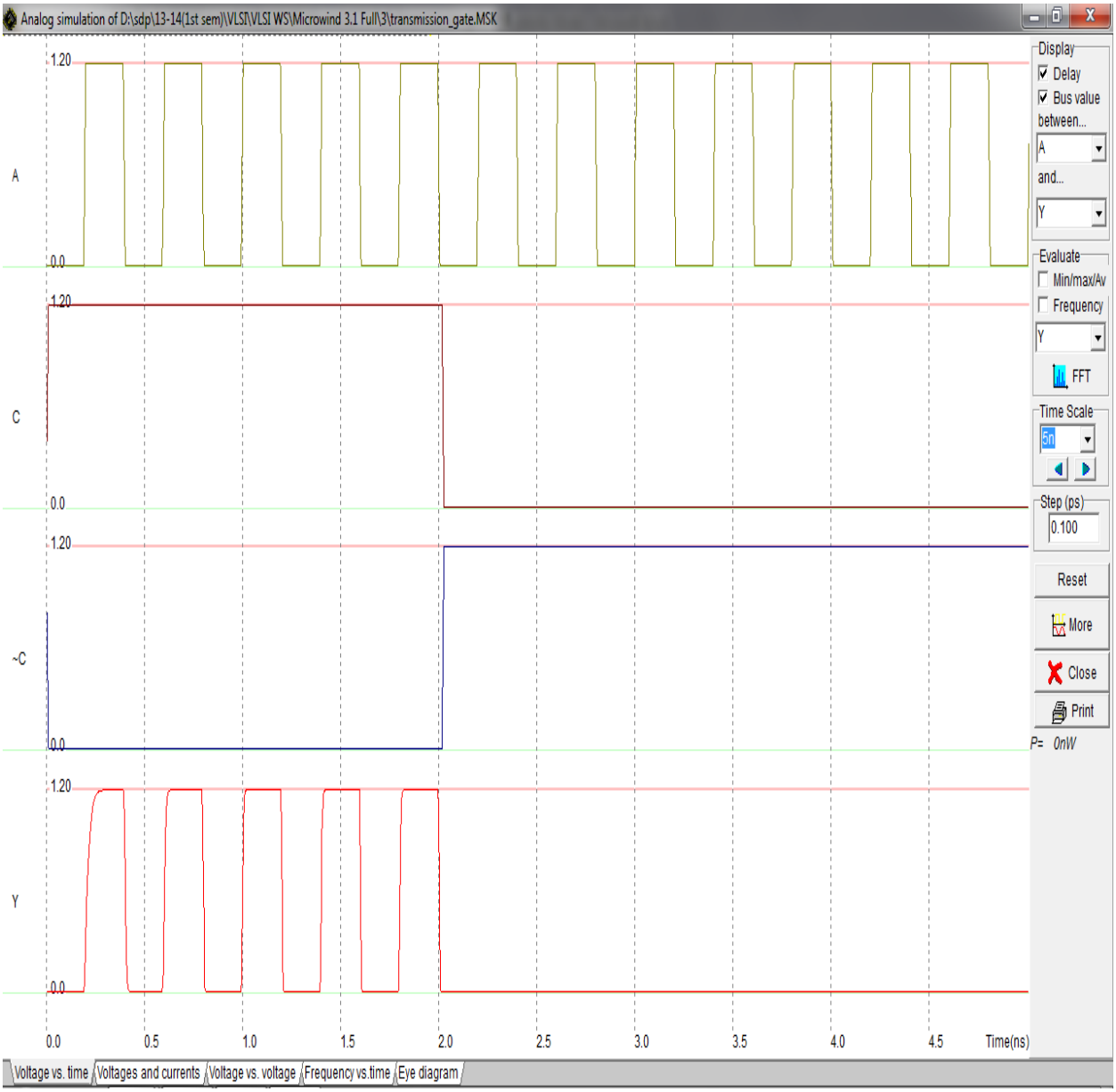




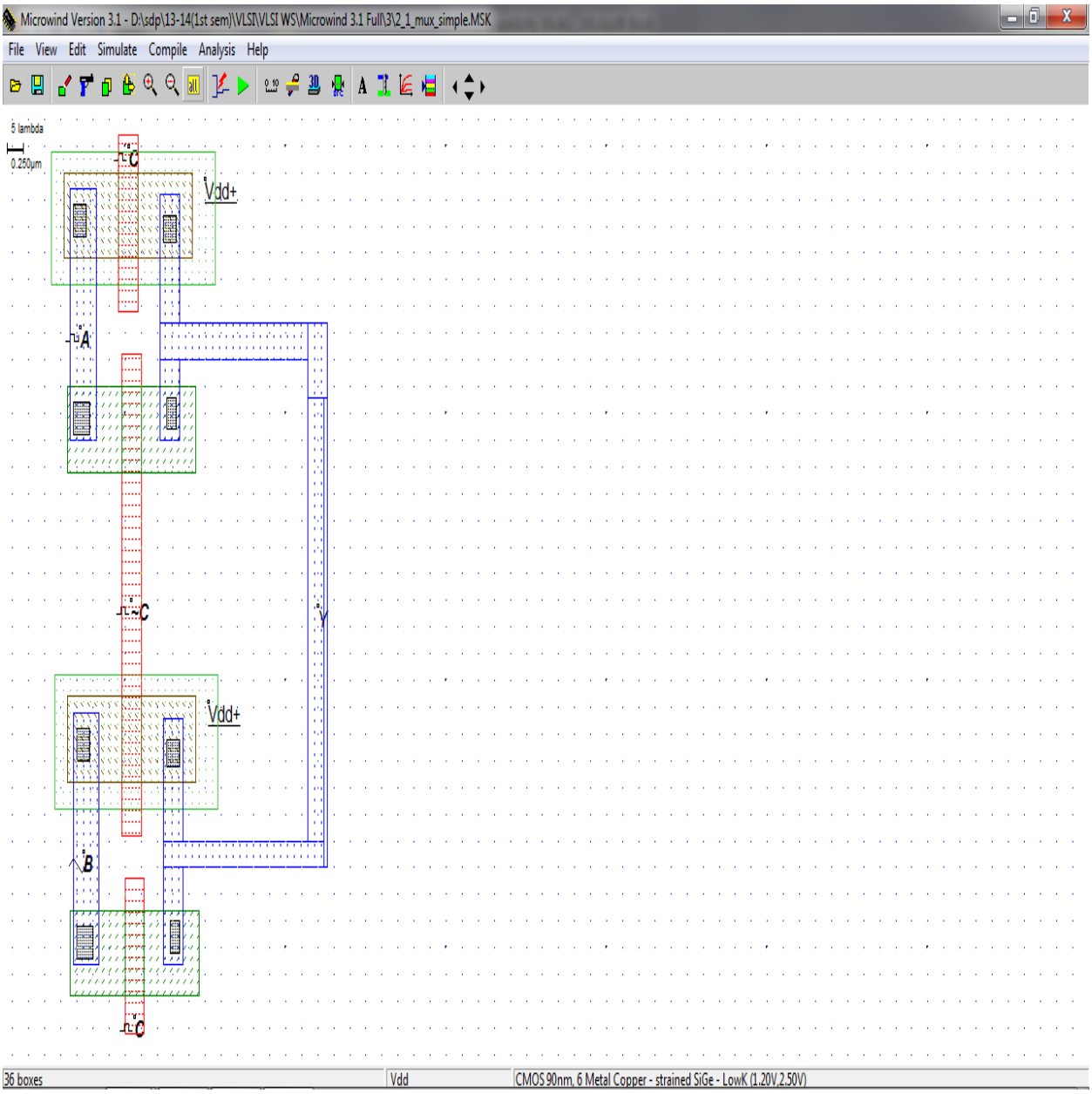
Simple transmission gate:



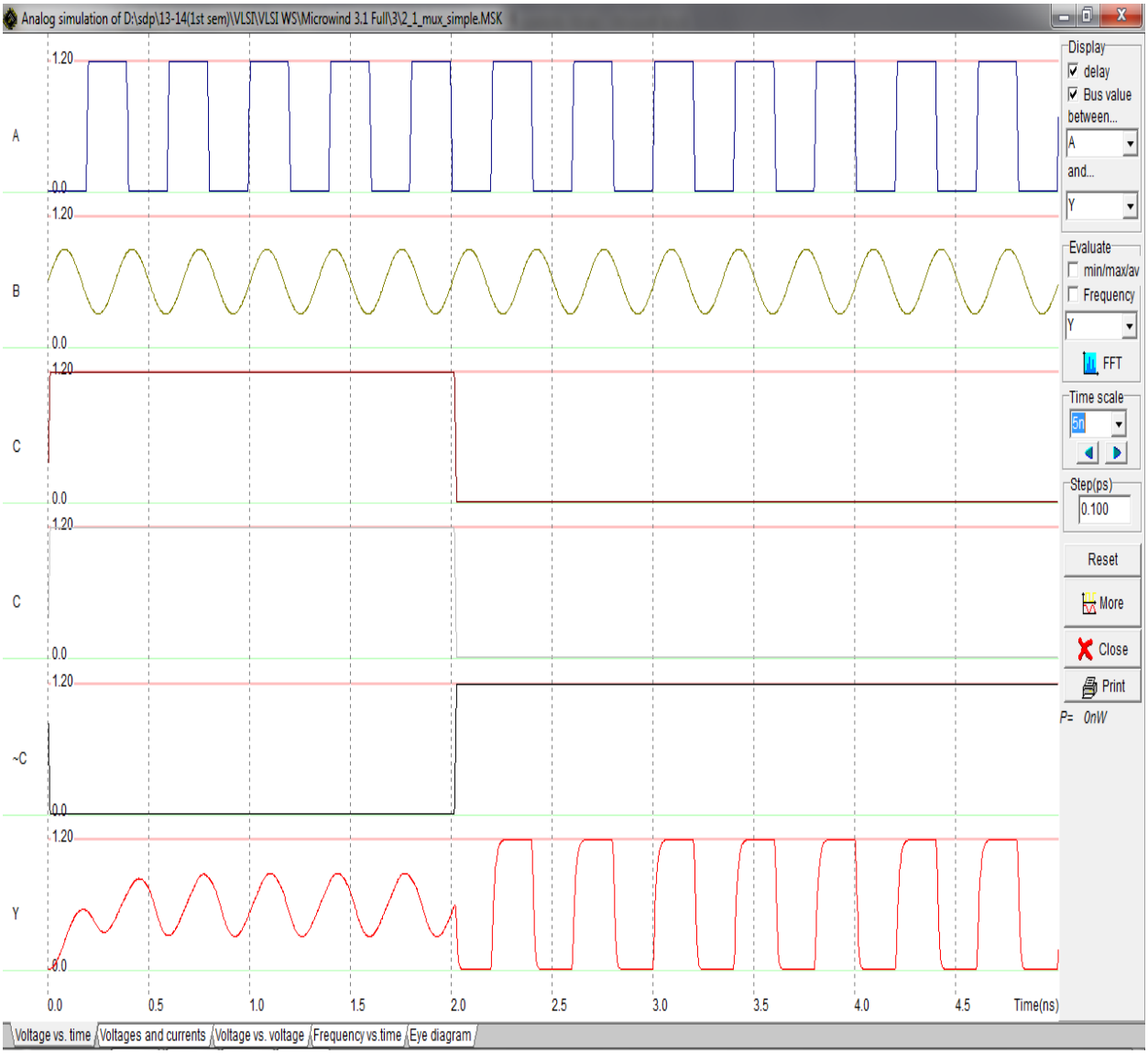
Waveform:


Analog simulation of D:\sdp\13-14(1st sem)\VLSI\VLSI WS\Microwind 3.1 Full\3\transmission_gate.MSK
A
C
~C
Y
1.20
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Time(ns)
Display
Delay
Bus value
between...
A
and...
Y
Evaluate
Min/max/Av
Frequency
Y
FFT
Time Scale
5n
Step (ps)
0.100
Reset
More
Close
Print
P= 0nW
Voltage vs. time
Voltages and currents
Voltage vs. voltage
Frequency vs.time
Eye diagram

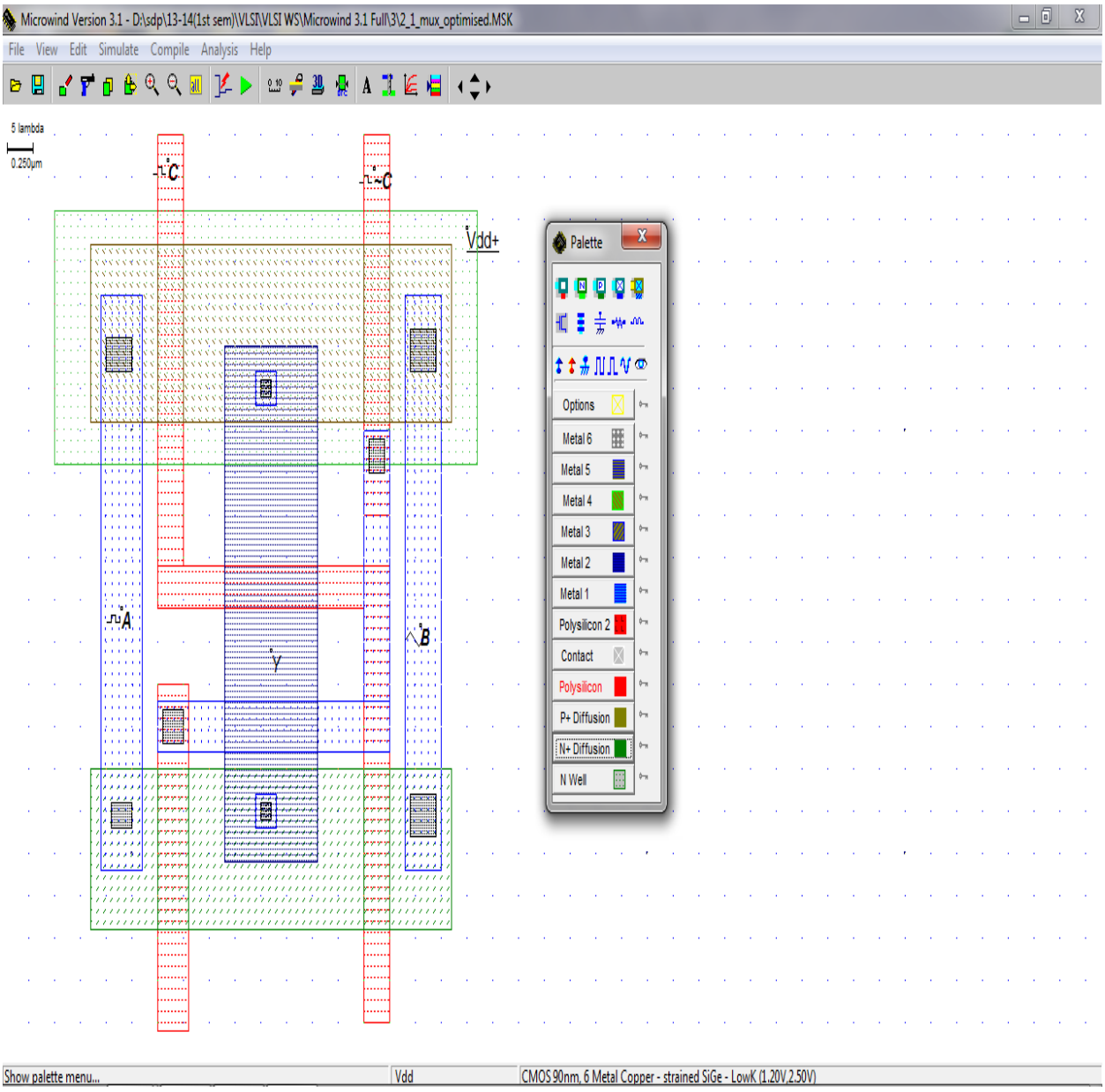
2:1 MUX using transmission gate (1)



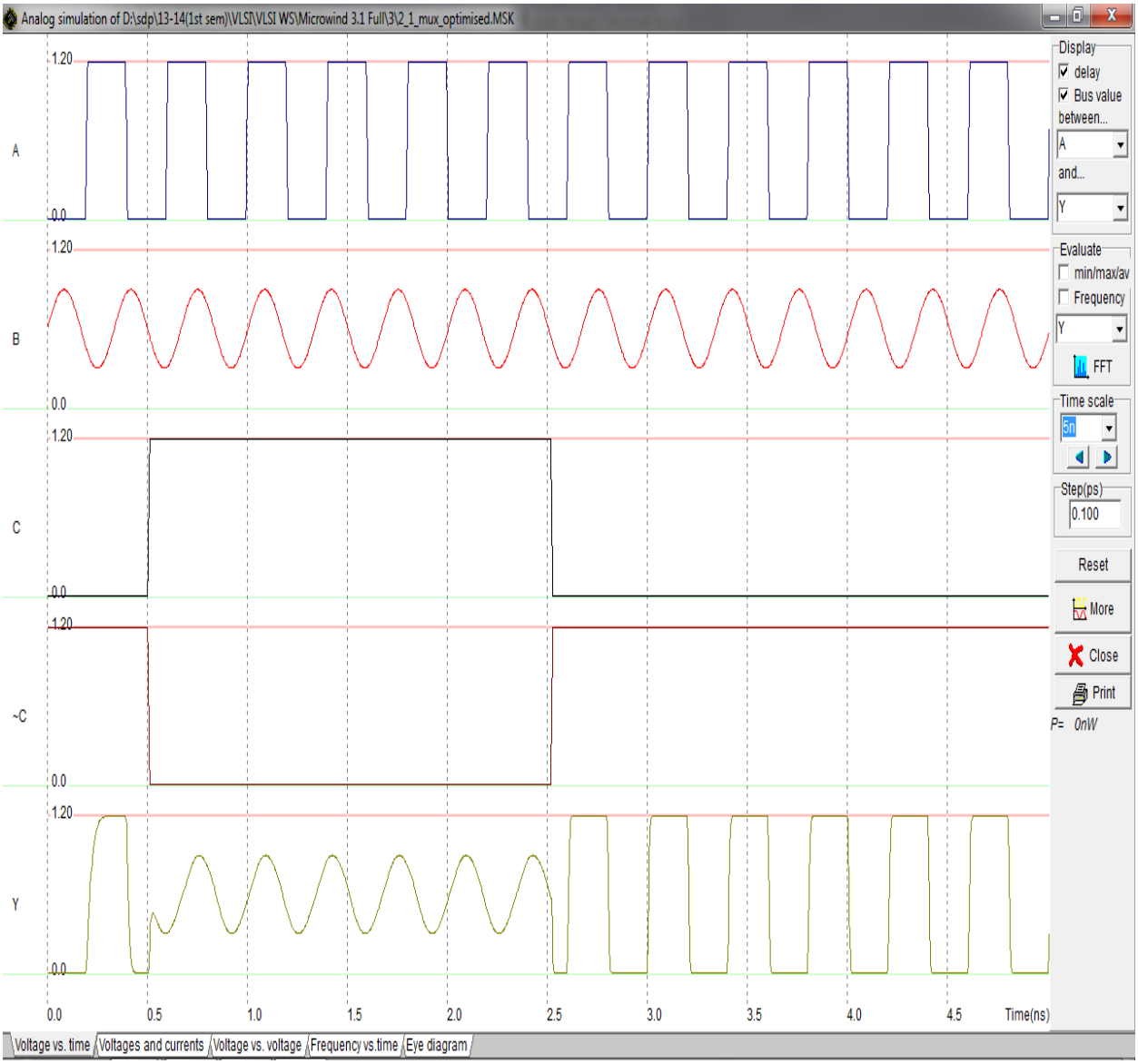
Waveform:


Analog simulation of D:\sdp\13-14(1st sem)\VLSI\VLSI WS\Microwind 3.1 Full\3\2_1_mux_simple.MSK
A
B
C
C
~C
Y
1.20
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Time(ns)
Display
delay
Bus value
between...
A
and...
Y
Evaluate
min/max/av
Frequency
Y
FFT
Time scale
5n
Step(ps)
0.100
Reset
More
Close
Print
P= 0nW
Voltage vs. time
Voltages and currents
Voltage vs. voltage
Frequency vs.time
Eye diagram

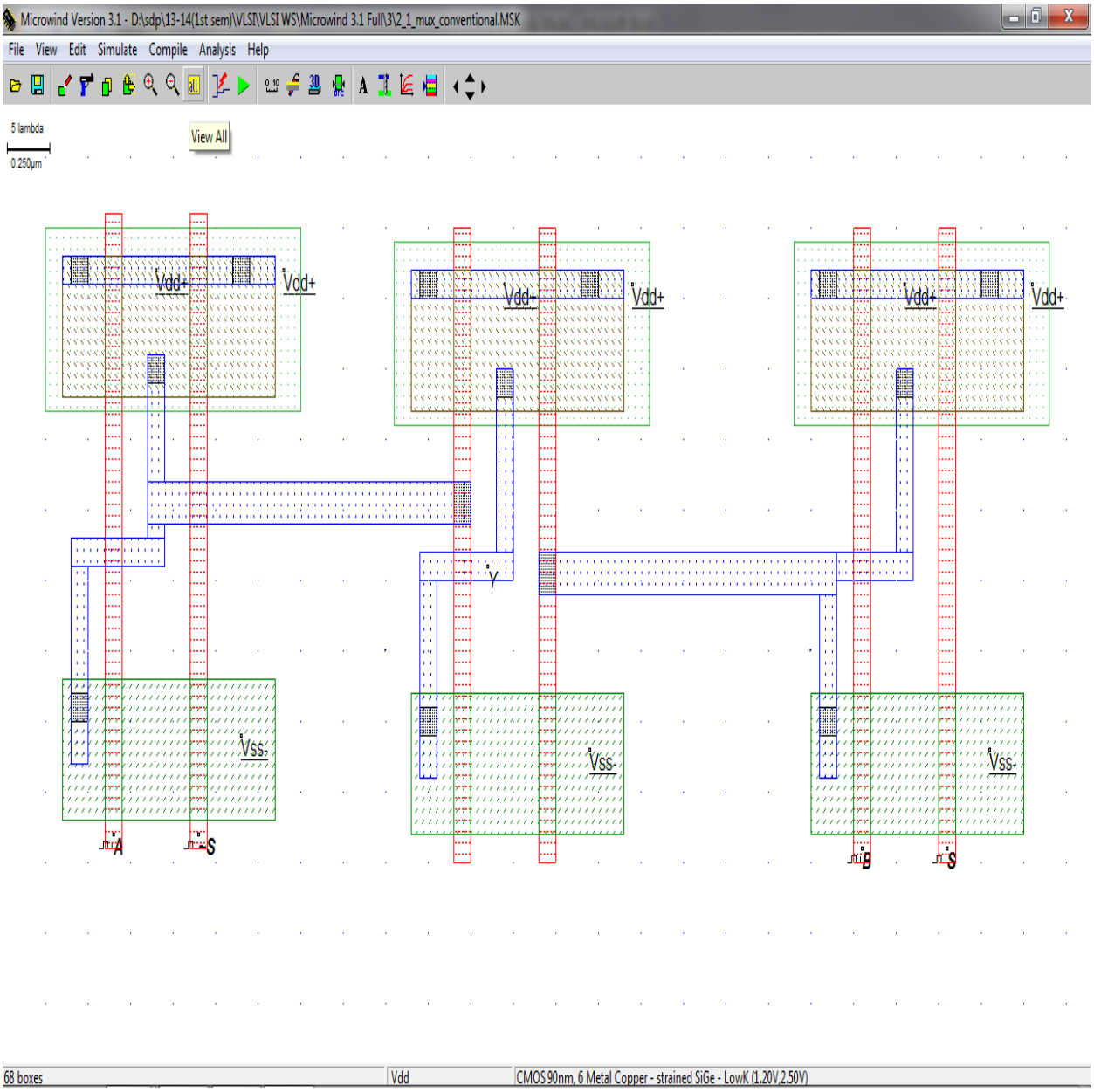
2:1 MUX using transmission gate method (2):



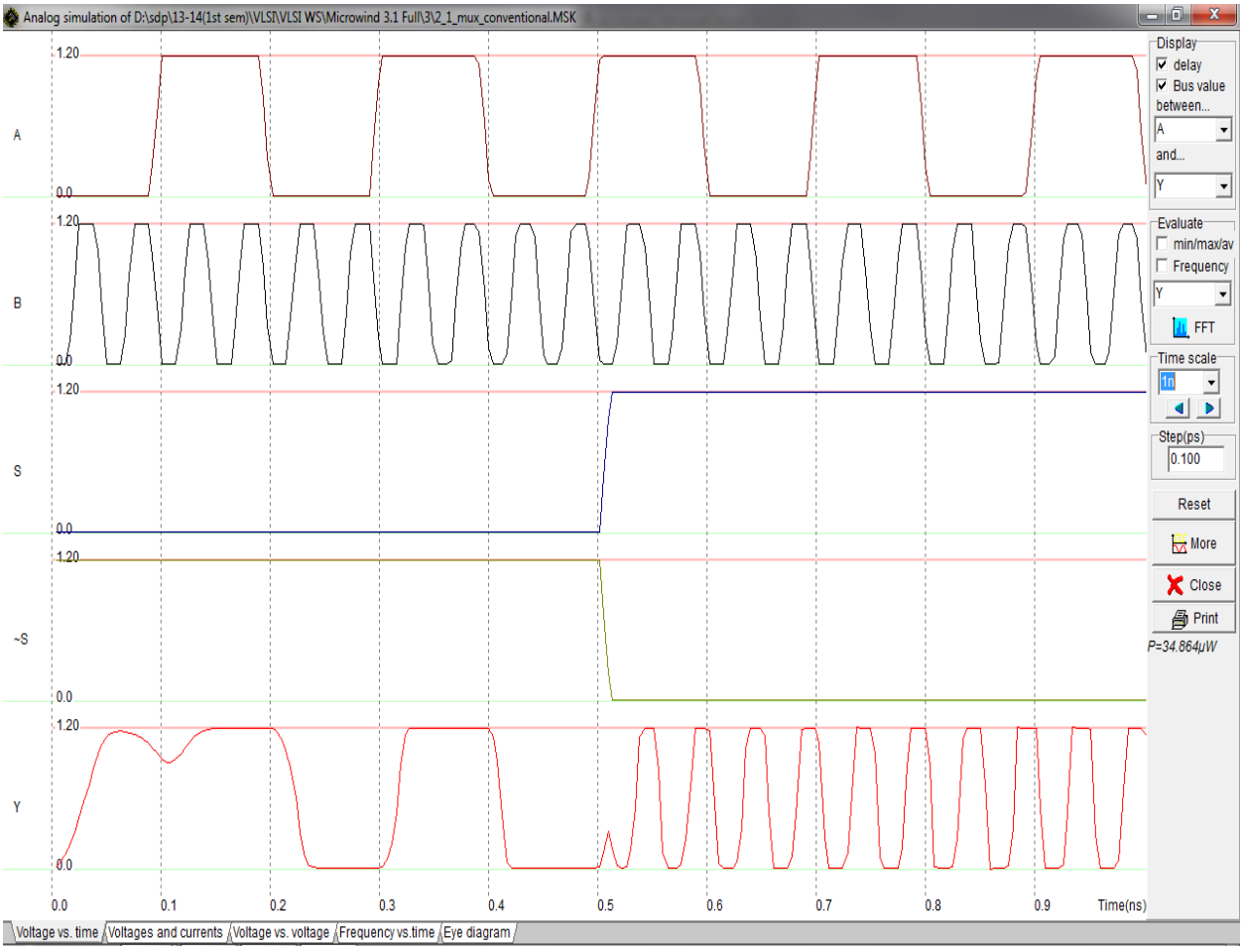
Waveform:


Analog simulation of D:\sdp\13-14(1st sem)\VLSI\VLSI WS\Microwind 3.1 Full\3\2_1_mux_optimised.MSK
A
B
C
~C
Y
1.20
0.0
Display
delay
Bus value
between...
and...
Evaluate
min/max/av
Frequency
FFT
Time scale
5n
Step(ps)
0.100
Reset
More
Close
Print
P= 0nW
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Time(ns)
Voltage vs. time
Voltages and currents
Voltage vs. voltage
Frequency vs.time
Eye diagram

2:1 MUX using Conventional method:



Waveform:


Analog simulation of D:\sdp\13-14(1st sem)\VLSI\VLSI WS\Microwind 3.1 Full\3\2_1_mux_conventional.MSK
A
B
S
~S
Y
1.20
0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Time(ns)
Display
delay
Bus value
between...
A
and...
Y
Evaluate
min/max/av
Frequency
Y
FFT
Time scale
1n
Step(ps)
0.100
Reset
More
Close
Print
P=34.864µW
Voltage vs. time
Voltages and currents
Voltage vs. voltage
Frequency vs.time
Eye diagram

Conclusion: -

In this practical we can see that for the same logic TG consumes less area compared to conventional method.

Questions:

Q1. Give the details of power dissipations in CMOS.

Q.2.Explain channel length Modulation of MOSFET in detailed.

Q.3.Explain Body effect in MOSFET.

Q.4. Give the details of delays in CMOS.

Reference Book:

Neil H.E.West, David Money Harris, “CMOS VLSI Design(A Circuits and System perspective)”, Fourth Edition, Pearson Education, Inc., publishing as Addison-Wesley, ch.1, pp-12-15